



## RĪGAS TEHNISKĀ UNIVERSITĀTE

Reģ.Nr.9000068977, Kipsalas iela 6A, Rīga, LV-1048, Latvija  
Tālr.:67089999; Fakss:67089710, e-pasts:rtu@rtu.lv, www.rtu.lvwww.rtu.lv

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### Study programme "Business Informatics"

#### Main attributes

|                                            |                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                                      | Business Informatics                                                                                                                                                                                                                                                                                                                                                       |
| Identification code                        | DMB0                                                                                                                                                                                                                                                                                                                                                                       |
| Education classification code              | 45483                                                                                                                                                                                                                                                                                                                                                                      |
| Level and type                             | Second-Cycle Higher Education (Academic Master) Studies                                                                                                                                                                                                                                                                                                                    |
| Higher education study field               | Information Technology, Computer Engineering, Electronics, Telecommunications, Computer Control and Computer Science                                                                                                                                                                                                                                                       |
| Head of the study field                    | Agris Nikitenko                                                                                                                                                                                                                                                                                                                                                            |
| Deputy head of the study field             | Jurgis Poriņš                                                                                                                                                                                                                                                                                                                                                              |
| Department responsible                     | Faculty of Computer Science, Information Technology and Energy                                                                                                                                                                                                                                                                                                             |
| Head of the study programme                | Mārite Kirikova                                                                                                                                                                                                                                                                                                                                                            |
| Professional classification code           | -                                                                                                                                                                                                                                                                                                                                                                          |
| The type of study programme                | Full time, Part time                                                                                                                                                                                                                                                                                                                                                       |
| Language                                   | Latvian, English                                                                                                                                                                                                                                                                                                                                                           |
| Accreditation                              | 29.11.2023 - 30.11.2029; Accreditation certificate No 2023/44-A                                                                                                                                                                                                                                                                                                            |
| Volume (credit points)                     | 120.0                                                                                                                                                                                                                                                                                                                                                                      |
| Duration of studies (years)                | Full time studies - 2.0; Part time studies - 2.5                                                                                                                                                                                                                                                                                                                           |
| Degree or/and qualification to be obtained | Master's degree of engineering science in business informatics                                                                                                                                                                                                                                                                                                             |
| Qualification level to be obtained         | The 7th level of European Qualifications Framework (EQF) and Latvian Qualifications Framework (LQF)                                                                                                                                                                                                                                                                        |
| Programme prerequisites                    | First cycle higher education in the field of computer science and information technology, or comparable education; or first cycle higher education in another field and an entrance exam or at least two years of professional experience in a field related to computer science and information technology. English language skills equivalent to at least CEFR B2 level. |

#### Description

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Abstract | <p>The study programme is an interdisciplinary programme that has been implemented since 2010/2011. study year. The study programme was developed in cooperation with the University of Buffalo (USA) and International Business Machines Corporation (IBM). The study language is English.</p> <p>In the study programme "Business Informatics" (total 120 ECTS), 57 ECTS are allocated for compulsory study courses, 27 ECTS to compulsory limited choice study courses, 6 ECTS to free choice, and 30 ECTS to the Master Thesis.</p> <p>There are three types of study courses: ICT study courses (for example, Advanced Data Technologies), business study courses (for example, Business Ethics) and integrated study courses (for example, Business Process Management and Engineering). Each study course has clearly defined learning outcomes and evaluation mechanisms. Various business and ICT issues are integrated in the study programme at the level of the study programme and at the level of individual study courses.</p> <p>Local and international students study together in the study programme. The enrolment to the study programme occurs in summer and in winter. Students of the summer intake have Master Thesis as a separate semester in their study plan to facilitate their mobility. Winter intake students can develop their Master Thesis more gradually.</p> |
| Aim      | <p>The aim of the study programme is to prepare professionals with expertise in systems thinking and engineering sciences who are able to use, choose, develop, and acquire ICT solutions that enable enterprise development; who are able to design intra- and inter-organizational information systems and are capable of participating in corresponding interdisciplinary and international projects.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Tasks    | <p>Tasks of the study programme:</p> <ul style="list-style-type: none"><li>- to develop students' systems thinking ability and their skills to use systems theory in constructing solutions which promote the development of businesses and science;</li><li>- to integrate business in ICT topics at different levels of granularity (inside the courses and among the courses);</li><li>- to apply the newest ICT developments in the study process, facilitate self-organized studies and technology-supported as well as traditional teamwork/group work;</li><li>- to assure the learning outcomes defined for the programme;</li><li>- to prepare students for their doctoral studies;</li><li>- to assure the flexibility of the study programme and possibility to modify it in order to follow changes in the labour market and new developments in different ICT and business areas;</li><li>- to develop cooperation with similar or topic-related programmes in other countries within ERASMUS and other agreements.</li></ul>                                                                                                                                                                                                                                                                                                                                                         |

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|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning outcomes                             | <p>Graduates of the study programme:</p> <ul style="list-style-type: none"> <li>- can identify business goals which are supportable by ICT solutions;</li> <li>- can identify business problems which are solvable by ICT solutions;</li> <li>- can, using appropriate technologies, model and analyse business processes, enterprise and business architecture, and information flows, as well as to design internal and inter-institutional information systems;</li> <li>- are able to follow advances concerning computer systems, communication technologies, and software and methods of their usage and to suggest various solutions and their combinations for raising competitiveness of enterprises and enterprise networks;</li> <li>- are able, using appropriate technologies, to develop enterprise improvement strategies, to plan analysis and change management projects, and define requirements for new products and services;</li> <li>- are able to interpret business concepts in computer science and ICT terms and vice versa;</li> <li>- can motivate, educate, and train employees to use the most appropriate ICT solutions, as well as participate in and lead inter-disciplinary and international teams;</li> <li>- are able to participate in international scientific projects in the area of business informatics as well as to propose and lead scientific projects;</li> <li>- are able to follow the rules of ethics in business and information systems development, and systems analysis.</li> </ul> |
| Final/state examination procedure, assessment | <p>Final examination procedure includes development of Master Thesis (30 ECTS). The Master Thesis is author's original research, where methods, models, techniques and prototypes applicable for solving tasks in the field of business informatics are analytically or experimentally assessed and/or integrated and/or designed. The purpose of the Master Thesis is to give students an opportunity to apply their knowledge and skills in the field of scientific research in order have firm grounding for post graduate studies; to further develop their competence in decision making, problem identification, analysis, and solving, as well as to promote creativity and sharpen professional discussion and presentation skills.</p> <p>The learning outcomes in Master Thesis are evaluated according to 10 grade (10 – the highest) system.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description of the future employment          | <p>Graduates of the study programme work for IT companies, public and private institutions using advanced ICT solutions, and consulting firms. They are qualified to fill the positions of Business Architect; Information Architect; Information Systems Architect; Information Engineer; Information Systems Engineer; Systems Analyst and Designer; Business Process Analyst, Engineer, and Manager; Business Analyst, Risk Manager, Requirements Engineer; Chief Information Officer; IS Manager; Business and IT Consultant, and the like.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Opportunity to continue studies               | <p>Opportunity to continue doctoral studies in Latvia, for instance, in the following study programmes:</p> <ul style="list-style-type: none"> <li>- Riga Technical University "Computer Science and Information Technology";</li> <li>- Riga Technical University "E-studies Technology and Management";</li> <li>- Riga Technical University "Environmental Engineering";</li> <li>- University of Latvia "Computer Science and Mathematics";</li> <li>- Vidzeme University of Applied Sciences "Socio-technical Systems Modelling".</li> </ul> <p>Students can continue their education in doctoral study programmes in the area of informatics in other EU countries.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Courses

| No        | Code                   | Name                                                                         | Credit points |
|-----------|------------------------|------------------------------------------------------------------------------|---------------|
| <b>A</b>  |                        | <b>Compulsory Study Courses</b>                                              | <b>57.0</b>   |
| 1         | <a href="#">DE0745</a> | Advanced data technologies                                                   | 6.0           |
| 2         | <a href="#">DE0747</a> | Enterprise Architecture and Requirements Engineering                         | 6.0           |
| 3         | <a href="#">DE0752</a> | Systems Theory                                                               | 6.0           |
| 4         | <a href="#">DE0740</a> | Business Process Management and Engineering                                  | 6.0           |
| 5         | <a href="#">DE0751</a> | e-Business Solutions                                                         | 6.0           |
| 6         | <a href="#">DE0645</a> | Portfolio Management Technologies                                            | 6.0           |
| 7         | <a href="#">DE0739</a> | Knowledge Management Systems                                                 | 6.0           |
| 8         | <a href="#">DE0754</a> | Service Science, Management, and Engineering                                 | 6.0           |
| 9         | <a href="#">DE0742</a> | Quality, Risk and Security Technologies                                      | 6.0           |
| 10        | <a href="#">DE0746</a> | Research Methods in Business Informatics                                     | 3.0           |
| <b>B</b>  |                        | <b>Compulsory Elective Study Courses</b>                                     | <b>27.0</b>   |
| <b>B1</b> |                        | <b>Field-Specific Study Courses</b>                                          | <b>21.0</b>   |
|           |                        | <i>Networking</i>                                                            | <i>3.0</i>    |
| 1         | <a href="#">DE0001</a> | Storage Networking                                                           | 3.0           |
| 2         | <a href="#">DE0748</a> | Network Security Requirements                                                | 3.0           |
|           |                        | <i>Specific Software Applications</i>                                        | <i>6.0</i>    |
| 1         | <a href="#">DE0743</a> | Artificial Intelligence in Business                                          | 6.0           |
| 2         | <a href="#">DE0753</a> | Information Systems Security Engineering                                     | 6.0           |
|           |                        | <i>Enterprise Information Systems</i>                                        | <i>6.0</i>    |
| 1         | <a href="#">DE0738</a> | Enterprise Information Technology Architecture, Applications and Integration | 6.0           |
| 2         | <a href="#">DE0638</a> | Digital Transformation                                                       | 6.0           |
|           |                        | <i>Analytics</i>                                                             | <i>6.0</i>    |
| 1         | <a href="#">DE0750</a> | Business Analytics                                                           | 6.0           |
| 2         | <a href="#">DE0940</a> | Business Analytics with Machine Learning                                     | 6.0           |
| <b>B2</b> |                        | <b>Humanities and Social Sciences Study Courses</b>                          | <b>6.0</b>    |
| 1         | <a href="#">BS0055</a> | Business Law                                                                 | 6.0           |
| 2         | <a href="#">BS0054</a> | Business Ethics                                                              | 6.0           |
| 3         | <a href="#">BS0052</a> | Entrepreneurship                                                             | 6.0           |
| 4         | <a href="#">BS0053</a> | Business Communication Skills                                                | 6.0           |
| 5         | <a href="#">BS0050</a> | Entrepreneurial Finance                                                      | 6.0           |
| 6         | <a href="#">BS0051</a> | Leadership                                                                   | 6.0           |
| 7         | <a href="#">BS0049</a> | Corporate Governance                                                         | 6.0           |
| 8         | <a href="#">DE0741</a> | Pedagogical Process. Basics                                                  | 6.0           |
| 9         | <a href="#">DE0643</a> | Pedagogy                                                                     | 3.0           |
| 10        | <a href="#">DE0729</a> | Communication and Presentation Skills                                        | 3.0           |
| <b>C</b>  |                        | <b>Free Elective Study Courses</b>                                           | <b>6.0</b>    |
| <b>E</b>  |                        | <b>Final Examination</b>                                                     | <b>30.0</b>   |
| 1         | <a href="#">DE0744</a> | Master Thesis                                                                | 30.0          |